



HFDC-4202 NT

Crosslinkable Power Cable Insulation Compound

HFDC-4202 NT is a long-life, water-tree-retardant, unfilled, crosslinkable, low density polyethylene insulation compound. The permanent tree-retardant additive provides improved performance in power cables in service involving exposure to moisture while retaining the excellent physical, electrical, and processing attributes of conventional, crosslinkable polyethylene.

HFDC-4202 NT compound provides electric utilities with:

- State-of-the-art “water” tree-retardant technology, consistently outperforming conventional XLPE in all accelerated cable wet aging tests at ambient and elevated temperatures
- Proven in the field tree retardance technology for over 25 years, at a broad cross section of geographical conditions, demonstrating excellent reliability and help achieving lowest life cycle costs
- Excellent electrical performance demonstrated by very high aged electrical strength and very low aged dissipation factor and power loss.

HFDC-4202 NT compound represents the state-of-the-art in tree-retardant, power cable insulation compounds.

Specifications

HFDC-4202 NT tree-retardant compound is designed for use in power distribution and sub-transmission cables, especially in underground applications. HFDC-4202 NT provides improved performance over XLPE cables and is recommended, with or without moisture barriers, for use as cable insulation up to and including 46 kV applications. Cables insulated with HFDC-4202 NT, using sound commercial fabrication manufacturing practice, would be expected to meet the latest editions of the following specifications and regulations:

- ANSI/ICEA: S-94-649, S-97-682, S-93-639 / NEMA WC74 (TR-XLPE requirements)
- AEIC: CS8
- RUS 50-70 (U-1)
- CEA: WCWG-01, WCWG-02
- UL 1072
- IEC: 60502, 60840
- CENELEC: HD 620 S1
- DIN VDE 0276-620
- BSI BS 6622
- GB/T 12706
- DL/T 1070



Physical Properties	Test Method ⁽¹⁾	Typical Value ⁽²⁾
Physical		
Density, 23°C	ASTM D 792	0.92 g/cm ³
Tensile Strength	ASTM D 638	3200 psi (22 MPa)
Tensile Elongation	ASTM D 638	530%
Tensile Strength Retention, Heat Aging		
30 days @ 121°C	ASTM D 638	> 75%
30 days @ 136°C		> 75%
14 days @ 150°C		>75%
Tensile Elongation Retention, Heat Aging		
30 days @ 121°C		> 75%
30 days @ 136°C		> 75%
14 days @ 150°C		>75%
Degree of Crosslinking, Extractables	ASTM D 2765A	<20%
Hot Creep/Hot Set @ 392°F (200°C)	ICEA T-28-562	<100%/<10%
Electrical		
Dielectric Constant, 60 Hz	ASTM D 150	2.3
Dissipation Factor, 60 Hz	ASTM D 150	0.00030
DC Volume Resistivity, 73°F (23°C)	ASTM D 257	>1x10 ¹⁵ ohm-cm
Dielectric Strength, 125 mils (3.175 mm)		
Short Time (S/T)	ASTM D 149	26 kV/mm (660 V/mil)
Step-by-Step (S/S)		23 kV/mm (590 V/mil)
Water-Tree Growth Rate ⁽³⁾	ASTM D 6097	10%
Relative Water Tree Size ⁽⁴⁾	ASTM D 6097	25 – 25%

(1)Tests are made in accordance with current ASTM or Dow Methods.

(2)Nominal property values representing tests on molded, stress-relieved slabs. Cure times were 15 minutes at 175°C. Values are typical, and not to be construed as specifications.

(3)Rate is relative to rate for Dow's standard XLPE, FREQ = 1 kHz.

(4)Size is relative to size for Dow's standard XLPE.

Processing Techniques

HFDC-4202 NT provides excellent surface finish and outstanding output rates over a broad range of conditions. For optimum results, melt extrusion temperatures in the range of 240 to 285°F (116 to 140°C) are recommended, although higher melt temperatures are possible on certain equipment with due care. Generally, a 60-40-20 screen pack is recommended. However, specific recommendations for processing conditions can be determined when the application and type of processing equipment are known.

Storage

The environment or conditions of storage greatly influences the recommended storage time. Storage under extreme conditions may affect the quality, processing, or performance of the product. Storage should be in



accordance with good manufacturing practices. If proper warehousing and storage temperatures [dry conditions, between 50°F and 86°F (10°C and 30°C) in temperature] are utilized, this product may be stored by the customer for up to one year. In general, the product should be used within two years of the date of manufacture. It is recommended that the practice of using the product on a first-in / first-out basis be established.



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Published January, 2010

